ISO/IEC 25010: The SQuaRE Product Quality Model Technical Report

1. Introduction to the SQuaRE Framework

As a central pillar of the Systems and software Quality Requirements and Evaluation (SQuaRE) series, ISO/IEC 25010 establishes the formal taxonomy for specifying and evaluating software quality. Within the SQuaRE framework, "Quality" is defined as the degree to which a system satisfies the stated and implied needs of its various stakeholders, thereby providing measurable value.

To ensure a comprehensive evaluation, the standard bifurcates the quality model into two distinct domains:

- **Quality in Use Model:** This model mandates five characteristics related to the outcomes of system interaction within specific contexts.
- **Product Quality Model:** This model governs the static and dynamic properties of the software and computer system across nine distinct characteristics. This report focuses exclusively on the Product Quality Model.

2. Evolution and Structural Standard Changes

The transition from the legacy ISO 9126 standard to ISO/IEC 25010 reflects the industry's need for more rigorous requirements. The 2023 revision represents the most current maturation of this model, notably increasing the characteristic count to nine.

Feature	ISO 9126 (2001)	ISO/IEC 25010:2023	Evolution Notes
Total Characteristics	6	9	Expanded to include modern technical requirements.
User Interaction	Usability	Interaction Capability	Shifted focus to the exchange of information.
Env. Adaptation	Portability	Flexibility	Renamed to emphasize adaptation capacity.
Security & Compatibility	Not Present	Included	Integrated into the 2011/2023 SQuaRE models.
Safety	Not Present	Included	**The 9th characteristic formally added in 2023.**

Note: While Security and Compatibility were integrated during the initial 2011 SQuaRE transition, Safety is the specific addition that brought the model from eight to nine characteristics in the 2023 revision.

3. Core Quality Model: The Nine Characteristics

The ISO/IEC 25010 Product Quality Model classifies software quality into nine high-level characteristics:

- **Functional Suitability**: The degree to which a product provides functions that meet stated and implied needs when used under specified conditions.
- **Performance Efficiency**: The performance of a system relative to the amount of resources used under stated conditions.
- **Compatibility**: The degree to which a product can exchange information with other components or perform functions while sharing a common environment and resources.
- **Interaction Capability**: The degree to which a system can be interacted with by specified users to achieve goals with effectiveness, efficiency, and satisfaction.
- **Reliability**: The degree to which a system performs specified functions under specified conditions for a specified period of time.
- **Security**: The degree to which a product protects information and data based on appropriate types and levels of authorization.
- **Maintainability**: The degree of effectiveness and efficiency with which a product or system can be modified by its intended maintainers.
- **Flexibility**: The degree to which a product can be adapted for different or evolving hardware, software, or usage environments.
- **Safety**: The degree to which a product or system avoids harm to people, business, software, property, or the environment.

4. Detailed Analysis: Functional Suitability

Functional Suitability mandates that the system's functional set aligns with stakeholder objectives. It is decomposed into the following sub-characteristics:

- **Functional Completeness**: The degree to which the set of functions covers all the specified tasks and intended users' objectives.
- **Functional Correctness**: The degree to which a product or system provides accurate results when used by intended users, providing results with the required precision.
- **Functional Appropriateness**: The degree to which the functions facilitate the accomplishment of specified tasks and objectives.

5. Detailed Analysis: Performance Efficiency

This characteristic dictates the system's ability to maintain performance within specified parameters relative to resource consumption.

- **Time behaviour**: The degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.
- **Resource utilization**: The degree to which the amounts and types of resources—including CPU, memory, storage, network devices, energy, and materials—meet requirements during execution.
- **Capacity**: The degree to which the maximum limits of a product or system parameter meet requirements.

6. Detailed Analysis: Compatibility

Compatibility requirements ensure the system operates harmoniously within shared infrastructures.

- **Co-existence**: The degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without having a detrimental impact on any other product.
- **Interoperability**: The degree to which a system, product, or component can exchange information with other products and mutually use the information that has been exchanged.

7. Detailed Analysis: Interaction Capability (Formerly Usability)

Interaction Capability focuses on the user interface and the user's ability to complete tasks in various contexts. This characteristic incorporates eight sub-characteristics:

- **Appropriateness recognizability**: The degree to which users can recognize whether a product or system is appropriate for their needs.
- **Learnability**: The degree to which the functions of a product or system can be learnt to be used by specified users within a specified amount of time.
- **Operability**: The degree to which a product or system has attributes that make it easy to operate and control.
- **User error protection**: The degree to which a system prevents users against operation errors.
- **User engagement**: The degree to which a user interface presents functions and information in an inviting and motivating manner (evolved from interface aesthetics).
- **Inclusivity**: The degree to which a product or system can be used by people of various backgrounds, including age, ability, culture, language, and gender.
- **User assistance**: The degree to which a product can be used by people with the widest range of characteristics and capabilities (including accessibility) to achieve specified goals in a specified context of use.
- **Self-descriptiveness**: The degree to which a product presents appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources.

8. Detailed Analysis: Reliability

The Reliability characteristic mandates the system's ability to maintain a defined level of performance over time.

- **Maturity**: The degree to which a system, product, or component meets needs for reliability under normal operation.
- **Availability**: The degree to which a system is operational and accessible when required for use.
- **Fault tolerance**: The degree to which a system, product, or component operates as intended despite the presence of hardware or software faults.
- **Recoverability**: The degree to which a product or system can recover the data directly affected and re-establish the desired state of the system in the event of an interruption or failure.

9. Detailed Analysis: Security

Security requirements govern the protection of information and data to ensure only authorized subjects or systems have access.

- **Confidentiality**: Ensuring that data is only accessible to those with authorized access.
- **Integrity**: The degree to which a system prevents unauthorized access to, or modification of, computer programs or data.
- **Non-repudiation**: The degree to which actions or events can be proven to have taken place, so that the events or actions cannot be denied later.
- **Accountability**: The degree to which the actions of a subject or resource can be traced uniquely to that entity.
- **Authenticity**: The degree to which the identity of a subject or resource can be proved to be the one claimed.

10. Detailed Analysis: Maintainability

Maintainability represents the efficiency with which a system can be modified by its maintainers to correct defects or adapt to new requirements.

- **Modularity**: The degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.
- **Reusability**: The degree to which an asset can be used in more than one system, or in building other assets.
- **Analysability**: The effectiveness and efficiency with which it is possible to assess the impact on a product of an intended change, or to diagnose deficiencies or causes of failures.
- **Modifiability**: The degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.
- **Testability**: The degree of effectiveness and efficiency with which test criteria can be established and tests can be performed to determine whether those criteria have been met.

11. Detailed Analysis: Flexibility (Formerly Portability)

Flexibility assesses the capacity for the system to be transferred across different environments.

- **Adaptability**: The degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software, or other usage environments.
- **Installability**: The degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.
- **Replaceability**: The degree to which a product can replace another specified software product for the same purpose in the same environment.

12. Detailed Analysis: Safety

Safety is defined as the degree to which a product or system avoids harm to people, business, software, property, or the environment in a specified context of use.

From an auditor's perspective, "harm to business" may manifest as catastrophic financial loss due to system logic errors, while "harm to software" includes critical data corruption or the permanent loss of system integrity that renders the environment unusable.

13. Implementation: Static Analysis and Certification

The practical application of ISO/IEC 25010 is reinforced by automated tools and formal certification bodies.

- **Static Analysis Support**: Tools for static code analysis are critical for enforcing the ISO 25010 model. These tools help identify defects and ensure compliance with rigorous coding standards such as **MISRA**, **CERT**, and **AUTOSAR**.
    - **Security**: Static analysis identifies structural vulnerabilities and deficiencies to proactively prevent malicious attacks.
    - **Maintainability**: These tools directly support **Analysability** by identifying code deficiencies and improve **Testability** by reducing the total number of necessary test cases through early defect detection.
- **Professional Certification**: Official ISO/IEC 25000 certificates are now granted by accredited laboratories. Organizations have recently obtained certifications specifically for **Maintainability** and **Functional Suitability**, providing third-party validation of software product quality.

14. Conclusion

The ISO/IEC 25010 model remains the cornerstone of modern product quality evaluation. By establishing a rigorous hierarchy of characteristics and sub-characteristics, it provides the essential terminology and structure required to specify, measure, and evaluate software. For the systems auditor and software architect alike, it ensures that every facet of a system—from its core functionality to its long-term safety and modularity—is evaluated against a globally recognized standard.